

WTW 158 Worksheet 11 on Vector Algebra

1. Given $\vec{a} = \langle 2, 2, 0 \rangle$, $\vec{b} = \langle -1, 1, 0 \rangle$ and $\vec{c} = \langle 0, 1, 1 \rangle$.
 - 1.1 Explain why $\vec{a}$ and $\vec{b}$ form two sides of a right-angled triangle.
 - 1.2 Find the angle between $\vec{b}$ and $\vec{c}$.
2. Given the vectors $\vec{a} = \langle 2, -1, 0 \rangle$ and $\vec{b} = \langle 0, 1, 2 \rangle$ find
 - 2.1 $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{a}$ and $\vec{b} \times \vec{b}$
 - 2.2 $\vec{a} \cdot (\vec{a} \times \vec{b})$
3. Given the points $P(3, 1, 4)$ and $Q(-2, -2, 3)$.
 - 3.1 Find the distance between the point P and the xy -plane.
 - 3.2 Find the distance between the point P and the x -axis.
 - 3.3 Find the vector $\overrightarrow{QP}$.
4. Given the vectors $\vec{a} = \langle 1, 1, 4 \rangle$ and $\vec{b} = \langle 0, 2, -3 \rangle$
 - 4.1 Find a unit vector in the direction of $\vec{a}$.
 - 4.2 Find a vector of length 5 in the opposite direction to $\vec{a}$.
5. Find the angle between the vectors $\vec{a} = \langle 0, 1, -1 \rangle$ and $\vec{b} = \langle -1, -1, 0 \rangle$.
6. The angle between two unit vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$. Find $\vec{a} \cdot \vec{b}$.
7. Apply the right hand rule to find:
 - 7.1 $\vec{i} \times \vec{j}$ 7.2 $\vec{k} \times \vec{i}$
8. Find s such that $\vec{a} = \langle 0, s, -1 \rangle$ and $\vec{b} = \langle -1, s, 1 \rangle$ are orthogonal.
9. Find the area of the parallelogram formed by $\vec{a} = \langle 0, 1, -1 \rangle$ and $\vec{b} = \langle -1, -1, 0 \rangle$ as two of the sides.
10. Say whether each of the following is true or not and explain why.
 - 10.1 $(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$ for all vectors $\vec{a}$ and $\vec{b}$.
 - 10.2 $(\vec{u} + \vec{v}) \times \vec{v} = (\vec{u} + \vec{v}) \times \vec{u}$ for all vectors $\vec{v}$ and $\vec{u}$
- 11.1 Find the scalar projection of $\vec{b} = \langle -1, 3, 2 \rangle$ onto $\vec{a} = \langle 3, 2, 2 \rangle$
- 11.2 Find the vector projection of $\vec{b}$ onto $\vec{a}$.

Answers

- 1.1 $\vec{a} \cdot \vec{b} = 0$
- 1.2 $\theta = \frac{\pi}{3}$
- 2.1 $\langle -2, -4, 2 \rangle$, $\langle 2, 4, -2 \rangle$, $\langle 0, 0, 0 \rangle$
- 2.2 0
- 3.1 4
- 3.2 $\sqrt{17}$
- 3.3 $\langle 5, 3, 1 \rangle$
- 4.1 $\left\langle \frac{1}{\sqrt{18}}, \frac{1}{\sqrt{18}}, \frac{4}{\sqrt{18}} \right\rangle$
- 4.2 $\left\langle -\frac{5}{\sqrt{18}}, -\frac{5}{\sqrt{18}}, -\frac{20}{\sqrt{18}} \right\rangle$
5. $\frac{2\pi}{3}$
6. $\frac{1}{\sqrt{2}}$
- 7.1. $\vec{k}$ 7.2 $\vec{j}$
8. $s = \pm 1$
9. $\sqrt{3}$
- 10.1 Yes, $\vec{a} \times \vec{b} \perp \vec{b}$
- 10.2 No, LHS and RHS have opposite signs

11.1 $\frac{7}{\sqrt{17}}$ 11.2 $\frac{7}{17} \langle 3, 2, 2 \rangle$

Worksheet 11 on Vectors.

1.1 $\vec{a} \cdot \vec{b} = -2 + 2 = 0$

$\Rightarrow \vec{a} \perp \vec{b}$



1.2 $\cos \theta = \frac{\vec{b} \cdot \vec{c}}{|\vec{b}| |\vec{c}|}$
 $= \frac{1}{\sqrt{2} \cdot \sqrt{2}}$

$= \frac{1}{2}$

$\Rightarrow \theta = \pi/3$

2.1 $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 0 \\ 0 & 1 & 2 \end{vmatrix} = \langle -2, -4, 2 \rangle$

$\vec{b} \times \vec{a} = -\vec{a} \times \vec{b} = \langle 2, 4, -2 \rangle$

$\vec{b} \times \vec{b} = \vec{0}$

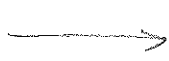
2.2 $\vec{a} \cdot (\vec{a} \times \vec{b})$

$= \langle 2, -1, 0 \rangle \cdot \langle -2, -4, 2 \rangle$

$= -4 + 4 = 0$

3.1 4

3.2 $\sqrt{17}$



3.3 $\vec{QP} = \langle 3 - (-2), 1 - (-2), 4 - 3 \rangle$
 $= \langle 5, 3, 1 \rangle$

4.1 $\frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{18}} \langle 1, 1, 4 \rangle$
 $= \langle \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{4}{3\sqrt{2}} \rangle$

4.2 $\langle \frac{-5}{3\sqrt{2}}, \frac{-5}{3\sqrt{2}}, \frac{-20}{3\sqrt{2}} \rangle$

5. $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$
 $= \frac{-1}{\sqrt{2} \cdot \sqrt{2}}$
 $= -\frac{1}{2}$
 $\Rightarrow \theta = \frac{2\pi}{3}$

6. $\cos \frac{\pi}{4} = \frac{\vec{a} \cdot \vec{b}}{1 \cdot 1}$

$\Rightarrow \vec{a} \cdot \vec{b} = \frac{1}{\sqrt{2}}$

7.1 $\vec{k}$

7.2 $\vec{j}$

8. $\langle 0, s, -1 \rangle \cdot \langle -1, s, 1 \rangle = 0$

$\Rightarrow s^2 - 1 = 0 \Rightarrow s = \pm 1$

9. $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & -1 \\ -1 & -1 & 0 \end{vmatrix}$
 $= \langle -1, 1, 1 \rangle$

$|\vec{a} \times \vec{b}| = \sqrt{3}$

10.1 True, $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$

10.2 Not true

LHS = $\vec{u} \times \vec{v} + \vec{v} \times \vec{u} = \vec{u} \times \vec{v}$

RHS = $\vec{u} \times \vec{u} + \vec{v} \times \vec{v} = \vec{v} \times \vec{u}$

LHS $\neq$ RHS

11.1 $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{7}{\sqrt{17}}$

11.2 $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \cdot \vec{a} = \frac{7}{17} \langle 3, 2, 2 \rangle$